

Erik Helmers

Fluent in English and Spanish

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EDUCATION

Bachelor of Science in Computer Science
Gonzaga University | Spokane, WA

September 2026 - May 2030

TECHNICAL SKILLS

Unreal Engine, Unity, C, C++, C#, HTML, JetBrains IDEs, Python, Git, GitHub, Windows, Debian 13, macOS, Android, WebGL, Arduino, Algorithms, Data structures, Adobe Premiere Pro (Certiport Certification)

ENGINEERING EXPERIENCE

PATHFINDER - Handheld Field Multi-Tool Firmware

Sep/2026- Oct/2026

Gonzaga University First Year Design Challenge | Spokane, WA

- Designed and deployed complete embedded firmware for a rugged handheld environmental sensor device integrating 10+ sensors across three I2C buses and UART interfaces.
- Implemented six cross-sensor integrations demonstrating practical signal processing and sensor fusion, directly supporting project judging criteria.
- Engineered robust sensor drivers with auto-detection, mock fallbacks, and graceful degradation.
- Optimized code efficiency to 50% flash (13.6 KB RAM) on a constrained platform, using fixed buffers, no heap allocation after startup, and integer math where floating-point is unavailable in the standard library.

Grog N' Gold - Lead Programmer

March/2026- May/2026

Digipen Academy | Redmond, WA | digipen-academy-wanic.itch.io/grog-and-gold

- Architected a comprehensive ship-based adventure game in Unity with 51 custom C# scripts, implementing enemy AI, a multi-tiered progression system with currency and shop mechanics, and a boss-encounter framework with weak-point detection.
- Engineered game state management, including a pause menu, scene switching, save/load functionality, and end-game tracking; optimized performance with 35+ animation sequences and spatial audio systems.
- Deployed fully functional builds demonstrating complex game systems and mechanics with high replayability through biome selection and player agency.

Scally & Wag: One Last Adventure - Lead Programmer & Game Director

March/2025- May/2025

Digipen Academy | Redmond, WA | digipen-academy-wanic.itch.io/2025scallyandwag

- Developed a multi-platform 2D arcade game in Unity using C# with water-themed gameplay, obstacle generation, health tracking, and collision detection across Android, WebGL, and Windows.
- Engineered advanced audio management with ambient sound, dynamic music triggers, and combo feedback; designed a custom camera controller with shake effects and parallax scrolling for visual depth.
- Built complete UI framework with high score tracking, main menu navigation, credits systems, and responsive interactions across multiple resolutions.

LEADERSHIP & SERVICE EXPERIENCE

GU Baja SAE "Holy Rollers"

Sep/2026– Present

Gonzaga University | Spokane, WA

- Initiating and architecting a telemetry system for the competition vehicle, designing sensor integration and a data acquisition framework to capture real-time performance metrics.

Varsity Team Captain

Sep/2024– May/2026

Newport High School | Bellevue, WA

- Led competitive Lacrosse and Water Polo student-athletes through strategic game planning, practice organization, and team morale initiatives, fostering accountability and growth.
- Coordinated directly with coaching staff to execute training regimens and manage in-game strategy adjustments.